DOMAIN 1, STUDY MATERIAL QUESTIONS

Question 1

Question: Placing a shield between an ionizing radiation source and an employee is an example of:

- a. Engineering control
- b. Administrative or management control
- c. Management control
- d. Only (a) and (b) above

Correct Answer: Select (a) **Explanation:** Hazard control is accomplished by either engineering or administrative (management) control. Generally, any type of control that requires design, construction, or design modifications is categorized as engineering control.

The best way to eliminate hazards is by means of engineering controls. However, if engineering controls cannot be implemented, then hazards must be controlled by means of administrative controls. Examples of administrative controls include training and reducing the time a worker is exposed to hazards.

Question 2

Question: A stock insurance company is best defined by which of the following statements?

- a. Insurance company which is profit motivated
- b. Insurance company with at least 100,000 shares of stock
- c. Insurance company protected by FDIC
- d. Insurance company that is not allowed to sell Workers' Compensation insurance

Correct Answer: Select (a) **Explanation:** Stock insurance companies are profit oriented, and are generally publicly traded firms. The company issues shares that can be purchased by the public in order to raise capital. Stock insurance companies pay their shareholders in capital appreciation, dividends, or both.

Question 3

Question: The BEST method to reduce employee exposure to airborne chemicals to an acceptable level is by means of:

- a. Administrative controls
- b. Using a respirator
- c. Reducing the number of exposure hours
- d. Engineering controls

Correct Answer: Select (d) **Explanation:** In control of any workplace hazard, engineering control must be considered first. Administrative controls such as job rotation and use of appropriate PPE must only be considered when engineering controls are either not feasible or cannot solve the problem.

Question 4

Question: Point source airborne contaminants are best controlled by:

- a. Dilution ventilation
- b. Local exhaust ventilation
- c. General ventilation
- d. General ventilation with a compression ratio of 10 or more

Correct Answer: Select (b) **Explanation:** Local exhaust ventilation is the most effective method of controlling point source airborne contaminants.

Dilution ventilation involves bringing in clean air to dilute the contaminated air, and then exhausting the diluted air. Dilution ventilation does not eliminate exposure, but should be designed in such a way as to bring the concentration down to a safe level.

Question 5

Question: Which of the following is an example of engineering noise control?

- a. Use of earmuffs
- b. Reducing the number of hours of exposure
- c. Use of sound absorbing materials
- d. Audiometric testing

Correct Answer: Select (c) **Explanation:** Engineering controls are used to permanently eliminate or lessen the severity of a hazardous condition. In this case, use of sound absorbing materials for control of over-exposure to noise is a good example of engineering control.

Question 6

Question: Which of the following insurance coverage is used to provide liability coverage for a company engaged in the cleanup of a hazardous waste site?

- a. Premium and medical coverage
- b. Chemical exposure coverage
- c. Manufacturers' and contractors' coverage
- d. Strict liability coverage

Correct Answer: Select (c) **Explanation:** This type of insurance coverage is for work that is performed away from the company, such as that done by consultants, contractors, and remediation companies.

Question 7

Question: The preferred angle range for fixed industrial stairs is:

- a. 30 to 35 degrees
- b. 15 to 25 degrees
- c. 10 to 25 degrees
- d. 5 to 15 degrees

Correct Answer: Select (a) **Explanation:** The preferred angle range for fixed industrial stairs is 30 to 35 degrees. The preferred angle for a ramp is no more than 15 degrees. The preferred angle range for fixed ladders is 75 to 90 degrees.

Question 8

Question: In ventilation design, the number of room air changes per unit time can be obtained by:

- a. Multiplying the room volume by the volumetric flow rate of air in the ventilation system.
- b. Dividing the room volume by the volumetric flow rate of air in the ventilation system.
- c. Dividing volumetric flow rate of air in the ventilation system by the room volume.
- d. Subtracting the volumetric flow rate of air in the ventilation system from the room volume.

Correct Answer: Select (c) **Explanation:** The number of air changes can be obtained by dividing the volumetric flow rate by the room volume. For example, a ventilation rate of 600 cubic feet per minute in a room of 300 cubic feet will provide 2 air changes per minute.

Question 9

Question: Which of the following organizations mandates lockout/tagout programs for service or maintenance of certain types of equipment?

- a. NIOSH
- b. OSHA
- c. FEMA
- d. CPSC

Correct Answer: Select (b) **Explanation:** The correct answer is OSHA.

Question 10

Question: Which of the following statements is NOT true in regard to chemical labels?

- a. Labels are used on drums of hazardous chemicals.
- b. Labels are used on transportation vehicles carrying hazardous materials.
- c. Labels are used on gas cylinders.
- d. Labels are used on bottles of hazardous chemicals.

Correct Answer: Select (b) **Explanation:** Placards are used on vehicles carrying hazardous materials. The Department of Transportation has standards and regulations (49CFR) for placards and labels.

Question 11

Question: In liability insurance coverage, the term "OCP" is an acronym for:

- a. Operation and Chemical Protection

- b. Owners and Contractors Protection
- c. Operators and Contractors Protection
- d. Operators and Constructors Protection

Correct Answer: Select (b) **Explanation:** "OCP" is an acronym for Owners and Contractors Protection. This type of insurance protects a business owner, property owner, or general contractor against liability due to the negligent act of an independent contractor or subcontractor who performs work on their behalf. The independent contractor or subcontractor purchases the policy. This type of policy covers a specific project at a specific location.

Question 12

Question: Insurance coverage falls into four broad categories. Which of the following is NOT one of these categories?

- a. Property insurance
- b. Liability insurance
- c. Life insurance
- d. Fire insurance

Correct Answer: Select (d) **Explanation:** Fire insurance is covered under property insurance. Health insurance is the fourth category.

Question 13

Question: Business Income Coverage (BIC) insurance addresses the loss exposures relating to loss of income and/or extra expense to operate. Which of the following is NOT one of the three elements of the loss exposure covered by the basic BIC form?

- a. Property
- b. Health
- c. Perils
- d. Consequences

Correct Answer: Select (b) **Explanation:** "Property" covers damage to or destruction of real property. "Perils" are covered causes of loss. "Consequences" covers loss of business income as a result of interruption or partial suspension of business.

Question 14

Question: Ground Fault Circuit Interrupters (GFCI) work on the principal of sensing a difference in the amount of electric current entering and leaving a circuit. The difference is considered to be electric current that is leaking to the ground. Once a difference has been detected, a GFCI opens the circuit in:

- a. 0.02 to 0.03 seconds
- b. 0.2 to 0.3 seconds
- c. 2 to 3 seconds
- d. 0.5 to 1.5 seconds

Correct Answer: Select (a) **Explanation:** GFCIs are fast-acting devices that, once a current leakage has been detected, open the circuit in 0.02 to 0.03 seconds.

A ground fault is an unintentional electrical path between a power source and a grounded surface. Ground faults most often occur when equipment is damaged or defective, such that live electrical parts are no longer adequately protected from unintended contact. If your body provides a path to the ground for this current, you could be burned, severely shocked, or electrocuted.

Question 15

Question: Which of the following fall protection systems absorb the energy of an actual fall?

- a. Fall arrest systems
- b. Fall depression systems
- c. Fall suppression systems
- d. Fall prevention systems

Correct Answer: Select (a) **Explanation:** Fall arrest systems are usually comprised of a lanyard that is connected to a beam through a choker. The lanyard is connected to an energy absorber that in turn is connected to a full body harness.

Question 16

Question: Ozone (O3) is a powerful oxidizing agent and is considered to be a hazardous gas. Which of the following operations is most likely to produce ozone?

- a. Paint spraying
- b. Electric arc welding
- c. Catalytic reforming
- d. Catalytic cracking

Correct Answer: Select (b) **Explanation:** Ozone is produced during electric arc welding operations.

Question 17

Question: Hexavalent chromium, which is toxic, can most likely be produced as a result of which of the following operations?

- a. Arc welding cast iron
- b. Distilling chromium oxide
- c. Welding stainless steel
- d. Precipitating hydrochromic salts

Correct Answer: Select (c) **Explanation:** Stainless steel contains chromium. Stainless steel can produce trivalent and hexavalent chromium. Hexavalent chromium is much more toxic than trivalent chromium. Toxic hexavalent chromium can be converted to trivalent chromium by a reduction chemical reaction.

Question 18

Question: The total pressure in a/an _____ ventilation system is equal to the sum of velocity pressure and static pressure.

- a. Open
- b. Atmospheric
- c. Dynamic
- d. Closed

Correct Answer: Select (d) **Explanation:** In closed ventilation systems, total pressure is equal to the sum of static and velocity pressure: TP = SP + VP.

Question 19

Question: Which of the following markings is used on many products manufactured in Europe, indicating that the product meets the health and safety standards?

- a. CE
- b. ES
- c. ESH
- d. EHS

Correct Answer: Select (a) **Explanation:** The CE marking on European products is similar to the UL marking used on many US products.

The CE marking is mandatory for products sold in the European Economic Area (EEA). By putting the CE marking on their products, manufacturers declare that their products meet the requirements set for by the European Commission (EC).

Question 20

Question: Per DOT regulations, _____ drivers are not allowed to consume alcohol within four hours before starting their driving shift.

- a. All
- b. Commercial
- c. Hazardous materials truck
- d. Commercial hazardous materials truck

Correct Answer: Select (b) **Explanation:** Per DOT regulations found in 49 CFR, commercial drivers are not allowed to consume alcohol within four hours prior to starting their driving shift.

Question 21

Question: The type of liability insurance that helps protect individuals and companies that provide professional advice and services from bearing the full cost of defending against a negligence claim made by a client is called:

- a. Professional liability insurance

- b. Professional indemnity insurance
- c. Errors and omissions insurance
- d. All of the above

Correct Answer: Select (d) **Explanation:** This type of coverage focuses on the policyholder's alleged failure to perform, cause financial loss, and errors and omissions in the service or product sold.

Question 22

Question: Which of the following statements is NOT true in regard to course of construction insurance?

- a. It covers a building under construction.
- b. It covers perils such as fire, wind, and theft.
- c. The policy will pay for damages up to the coverage limit.
- d. The coverage limit must accurately reflect the total value of the structure, including labor costs and land value.

Correct Answer: Select (d) **Explanation:** Course of construction insurance, also known as builders risk insurance, does not cover land value.

Question 23

Question: Per DOT regulations, the post-accident drug and alcohol test applies to all drivers who are involved in fatal crashes. The test must also be conducted on all CDL drivers who are cited for moving violations arising in a crash that requires a vehicle being towed or an injury requiring medical attention away from the scene. The alcohol test must be conducted within _____ hours and the controlled substances test must be conducted within _____ hours of the crash.

- a. 10, 25
- b. 8, 32
- c. 4, 8
- d. 15, 45

Correct Answer: Select (b) **Explanation:** CDL drivers are also subject to unannounced random testing for alcohol and drugs.

Question 24

Question: All of the following are components of an insurance contract EXCEPT:

- a. Exclusions
- b. Conditions
- c. Declarations
- d. Privity

Correct Answer: Select (d) **Explanation:** Privity refers to a one-on-one direct relationship and is not a component of an insurance contract.

Other components of an insurance contract include the following: • The declaration page lists the name of the insurance company, the name of the insured, what is covered under the policy, the amount of coverage, the time that the policy is in effect, and how much premium the insured must pay for the coverage. • The definitions section explains the terminology that is used in the policy. • The insurance agreement section states what the insurance company promises to do. • The exclusions section states what risks are not covered under the policy. • The conditions section specifies those provisions of the policy that are required to be met before the insurance company pays the claim. • The endorsements section contains amendments to the original policy.

Question 25

Question: Incorporation of safety at the design stage of a potentially hazardous industrial process, rather than after it is operating, is important because:

- a. OSHA regulations requires a place of employment free of all hazards.
- b. It has an economic advantage to the company.
- c. It decreases management liability before plant operation.
- d. It educates design engineers in safety standards.

Correct Answer: Select (b) **Explanation:** Incorporation of safety at the design stage has a clear economic advantage. It is much less expensive to make changes at this stage compared to changes made after construction of a facility.

Question 26

Question: In facility design, the responsibility for identifying and controlling hazards associated with equipment lies with:

- a. Mechanical engineers
- b. Purchasing agents
- c. Design team
- d. Equipment manufacturer

Correct Answer: Select (c) **Explanation:** The design team is responsible for understanding the hazards associated with equipment and control of those hazards.

Question 27

Question: A pump is sending cooling water to a reactor. Loss of cooling water to this reactor would cause the reactor to explode. A fail-safe design for the system is BEST portrayed by:

- a. Activation of a spare pump on failure of the cooling water pump.
- b. Automatic shut down of process on failure of the cooling water pump.
- c. Relieve pressure in the reactor.
- d. All of the above.

Correct Answer: Select (a) **Explanation:** The best fail-safe design is depicted in statement (a). This would allow the plant to operate safely and efficiently when the cooling water pump fails. The (b) statement would result in loss of production and additional cost for startup.

Question 28

Question: The concept of Energy Theory in safe design of processing facilities states that the severity of an accident is related to:

- a. Amount of energy release
- b. Rate of energy release
- c. Means of energy transfer
- d. All of the above

Correct Answer: Select (d) **Explanation:** The energy theory relates the kind and severity of injuries to statements (a), (b), and (c).

The energy theory states that the cause of accidents is energy transfer with enough force to cause bodily or property damage. Therefore, preventing and controlling energy buildup and creating an environment that is not capable of energy conduction can greatly reduce the potential for accidents.

Question 29

Question: Floor slipperiness is measured by:

- a. Slip meter
- b. Friction meter
- c. Viscometer
- d. Hydrometer

Correct Answer: Select (a) **Explanation:** A slip meter is a device that measures floor slipperiness. Slip meters measure the static coefficient of friction. The smaller the coefficient of friction, the more chance for people to slip on the floor.

Question 30

Question: One of the more commonly encountered hazards for ground supported scaffolds is:

- a. Breakage at a given leg
- b. Over-pressurizing the scaffold
- c. Breakage at the surface
- d. Tipping

Correct Answer: Select (d) **Explanation:** Tipping is one of the more commonly encountered hazards for ground supported scaffolds. This can be prevented by attaching the scaffold to a firm structure such as a building.

Question 31

Question: Drums containing Polychlorinated Biphenyls (PCB) must be made of:

- a. Metal
- b. Polyethylene
- c. Ceramic
- d. PVC

Correct Answer: Select (a) **Explanation:** PCB drums must be made of metal and must meet DOT specifications (49CFR).

Question 32

Question: The severity of electric shock primarily depends on:

- a. The electric current flowing through the body
- b. The body capacitance to absorb electrical energy
- c. The body transmittance for electrical conductivity
- d. All of the above

Correct Answer: Select (a) **Explanation:** The amount of electric current flowing through a victim's body determines the severity of an electric shock. Since current is dependent on the voltage and resistance of the body, these factors are also important considerations. For example, a circuit providing a potential difference of 220 volts sends more current through the body as compared to a circuit of 110 volts.

Question 33

Question: Ventricular fibrillation caused as a result of electric shock refers to:

- a. Blockage of the heart muscles
- b. Fibers of the heart contract separately
- c. Fibers of the heart contract simultaneously
- d. Fibers of the heart expand beyond limit of elasticity

Correct Answer: Select (b) **Explanation:** In order for the heart to perform its function as a pump, fibers of the heart must contract together in a coordinated manner. Ventricular fibrillation refers to the situation where these fibers contract separately resulting in cease of blood circulation in the body.

Question 34

Question: Interlocks used in electrical equipment must meet which of the following criteria?

- a. Lockout-tagout
- b. Fail-active
- c. Fail-passive
- d. Fail-safe

Correct Answer: Select (d) **Explanation:** Interlocks used in electrical equipment must be of the fail-safe type. This means that the device must be designed in such a way to compensate for its failure without creating a safety problem.

Question 35

Question: An electrical circuit protective device such as a fuse or circuit breaker protects the circuit from overload by:

- a. Closing the circuit
- b. Dissipating excess heat
- c. Opening the circuit
- d. Electrical bonding

Correct Answer: Select (c) **Explanation:** Fuses and other circuit protective devices automatically open the circuit in the event of overflow of current. This action stops the flow of current.

Question 36

Question: "Arcing" refers to:

- a. Flow of electrons through air
- b. Welding of metal parts
- c. Type of protective helmet
- d. Process of bending metal sheets

Correct Answer: Select (a) **Explanation:** "Arcing" is the flow of electrons through air between two conductors that are not in direct contact with each other. This electron flow generates a visible light.

Question 37

Question: Electrical bonding:

- a. Eliminates the electrical charges between two conductors
- b. Eliminates the potential difference between two conductors
- c. Sends electrical charges to the ground
- d. Reduces electrical power consumption

Correct Answer: Select (b) **Explanation:** Bonding does not remove the charges; however, it makes the potential difference between the two objects equal so there is no driving force for current flow between the two objects.

Question 38

Question: The electrolytic fluid in lead-acid batteries during charging can produce:

- a. Poisonous lead gas
- b. Lead oxide gas
- c. Hydrogen gas
- d. Lead hydride gas

Correct Answer: Select (c) **Explanation:** Hydrogen gas can be produced during charging of lead-acid batteries. If hydrogen reaches its flammable limits, there is a potential for explosion or fire.

Question 39

Question: An electrical system is considered to be de-energized (no potential difference in any component of the system):

- a. When the power to the circuit has been shut off
- b. After the system has been shut off and tested
- c. After the power has been shut off and proper lockout/tagout has been performed
- d. When it is grounded

Correct Answer: Select (b) **Explanation:** An electrical system can be considered de-energized only after appropriate tests have been conducted. For example, many electrical circuits having capacitors can store electrical energy for a long time after the system has been shut off. Also, systems that are grounded can have charged capacitors that might cause an electrical shock.

Question 40

Question: For control of electrostatic hazards which one of the following types of clothing is preferable?

- a. Non-conductive
- b. Semi-conductive
- c. Conductive
- d. Type of clothing has no effect

Correct Answer: Select (c) **Explanation:** Conductive clothing and shoes cause the electrostatic charges to safely flow to the ground.

Question 41

Question: The surface conductivity of an electrostatically charged surface:

- a. Will increase with an increase in humidity
- b. Will decrease with an increase in humidity
- c. Humidity has no effect on surface conductivity
- d. Will increase with an increase in temperature

Correct Answer: Select (a) **Explanation:** Humidity will increase surface conductivity. Humidity is the ratio of partial pressure of water vapor in air to equilibrium vapor pressure of water at a given temperature. An increase in temperature results in an increase in equilibrium vapor pressure and a decrease in relative humidity resulting in a decrease in surface conductivity.

Question 42

Question: Which of the following is associated with equipment failure for facility design and operation?

- a. Poor maintenance program

- b. Lack of fail-safe instrumentation
- c. Metal fatigue
- d. All of the above

Correct Answer: Select (d) **Explanation:** Statements (a), (b), and (c) are all associated with equipment failure.

Question 43

Question: Which of the following is NOT associated with plant layout for facility design and operation?

- a. Direction of wind
- b. Number of people required to operate the plant
- c. Cost of process control instruments
- d. Cost of future revision

Correct Answer: Select (c) **Explanation:** Plant layout does not take into account the cost of instrumentation.

Question 44

Question: In new chemical plant layout where most processing equipment is located outdoors, which of the items listed below should be considered first?

- a. To group similar equipment in one location
- b. To determine the direction of the prevailing wind
- c. To provide room for maintenance
- d. To minimize piping and pumping costs

Correct Answer: Select (b) **Explanation:** The direction of prevailing wind will determine the location of many pieces of equipment.

Question 45

Question: In designing outdoor plant layout, all equipment that may spill flammable materials should be located:

- a. In an isolated area on the upwind side
- b. On the downwind side
- c. Near fire stations and on the upwind side
- d. Close to the receiving area

Correct Answer: Select (b) **Explanation:** Flammables should be placed on the downwind side otherwise, it is possible that the wind would carry the vapors from a spill over the plant where they could be ignited.

Question 46

Question: The areas for containment of relatively large spills of liquefied petroleum gases must always be constructed below grade because:

- a. These gases have a vapor density that is greater than one.
- b. A person on the grade level or above will not be asphyxiated.
- c. Spilled material can flow to these areas by gravity.
- d. Only (a) and (b) above.

Correct Answer: Select (d) **Explanation:** Gravity flow is not a reason, because liquefied petroleum gases (LPG's) turn into gaseous form once the pressure is reduced to atmospheric. Construction of containment areas below grade allows movement of emergency vehicles to and from the spill without creating additional hazards such as fire or explosion.

Question 47

Question: Which of the following is the leading cause of work related injuries and deaths?

- a. Electrical shock
- b. Falls and slips
- c. Motor vehicle accidents
- d. Exposure to toxic chemicals

Correct Answer: Select (c) **Explanation:** Motor vehicle accidents are the leading cause of work related injuries and deaths in the United States.

Question 48

Question: When a source emits light that reaches a given surface, the surface can:

- a. Absorb the light
- b. Transmit the light
- c. Reflect the light
- d. All of the above

Correct Answer: Select (d) **Explanation:** Light is a wave phenomenon and like other waves it can be absorbed, transmitted, or reflected by a surface.

Question 49

Question: The quantity of illumination is measured by which of the following instruments?

- a. Foot-candle meter
- b. Illuminometer
- c. Transmitter
- d. Light meter

Correct Answer: Select (d) **Explanation:** The instrument used for measurement of the quantity of light is called a light meter.

Question 50

Question: Which of the following organizations set forth standards for eye and face protection?

- a. CPSC
- b. FDA
- c. NIOSH
- d. ANSI

Correct Answer: Select (d) **Explanation:** The American National Standard Institute ANSI sets forth standards for eye and face protection (ANSI Z87.1). These standards have been adopted by OSHA.

Question 51

Question: Most personal devices used in fall protection:

- a. Prevent a person from falling
- b. Minimize the impact of a fall
- c. Interrupt falls already in progress
- d. Shield an individual from falling objects

Correct Answer: Select (c) **Explanation:** Although statements in (a) and (b) are true, most personal fall protection devices are designed to interrupt a fall after its occurrence.

Question 52

Question: All of the following are effective measures for fall protection EXCEPT:

- a. Remove slipping and tripping hazards
- b. Barriers
- c. Hard hat
- d. Guard rails

Correct Answer: Select (c) **Explanation:** Hard hats are personal protective equipment used exclusively for head protection.

Question 53

Question: Which one of the following is NOT an administrative control of hazards?

- a. Discipline an employee for unsafe work practices
- b. Provide periodic training
- c. Provide work rotation to reduce fatigue
- d. Provide personal protective equipment

Correct Answer: Select (d) **Explanation:** Personal protective equipment is part of the "engineering control" of hazards.

Question 54

Question: Before an employer can assign an employee to jobs requiring the use of a respirator, the employer:

- a. Must consult with employee and obtain his/her consent to use a respirator.
- b. Must test employee's reactions and performance while using a respirator in a non-hazardous atmosphere.
- c. Must obtain written permission from a physician on employee's ability to use a respirator.
- d. Document employee's performance while using a respirator for a period of no less than 10 days.

Correct Answer: Select (c) **Explanation:** OSHA requires that only able persons be assigned to jobs which require the use of a respirator. This determination must be made by a qualified physician who informs the employer of the results of examination in writing.

Question 55

Question: The stress generated in an object as a result of application of a force is equal to the:

- a. Value of the compression stress minus the tensile stress
- b. Value of the compression stress minus the shear stress
- c. Value of the compression stress divided by the shear stress
- d. Value of the force divided by the area upon which it acts

Correct Answer: Select (d) **Explanation:** Stress is defined as the value of force divided by the area.

Question 56

Question: The term "creep" is referred to as:

- a. The ability of a given material to resist tensile stress
- b. Slow but permanent deformation of a material under a given load
- c. The ability of a material to withstand shear stresses
- d. An individual with uncivilized, and at times, unusual behavior

Correct Answer: Select (b) **Explanation:** Creep is referred to as very slow but permanent deformation of a material that is under a given load.

Question 57

Question: Shoring is required for all the trenches more than 4 feet deep unless:

- a. They are sloped to an angle of repose
- b. The trench is in a stable rock
- c. A ladder is provided
- d. Only (a) and (b) above

Correct Answer: Select (d) **Explanation:** No shoring is required for situations stated in (a) and (b).

Question 58

Question: A trench is:

- a. A narrow excavation
- b. Deeper than it is wide
- c. Never wider than 15 feet
- d. All of the above

Correct Answer: Select (d) **Explanation:** Situations presented in statements a, b, and c are correct about trenches.

Some OSHA recommended safety tips for trenches are listed below: • Do not enter an unprotected trench! • Trench collapses cause dozens of fatalities and hundreds of injuries each year. • Trenches 5 feet deep or greater require a protective system. • Trenches 20 feet deep or greater require that the protective system be designed by a registered professional engineer.

Protective systems for trenches are the following: • Sloping protects workers by cutting back the trench wall at an angle inclined away from the excavation. • Shoring protects workers by installing aluminum hydraulic or other types of supports to prevent soil movement. • Shielding protects workers by using trench boxes or other types of supports to prevent soil cave-ins.

Question 59

Question: The guard on a grinding wheel must:

- a. Completely enclose the wheel
- b. Cover the wheel as completely as the nature of the work permits
- c. Be adjustable
- d. Only (b) and (c) above

Correct Answer: Select (d) **Explanation:** It is impossible to fully enclose the grinding wheel as some portion of the wheels must be exposed to perform grinding on the part. Guards on grinding wheels must be adjustable to provide adequate protection when the diameter of the wheel decreases.

Question 60

Question: Which of the following organizations has the most number of standards for unfired pressure vessels?

- a. ANSI
- b. NFPA
- c. ASME
- d. OSHA

Correct Answer: Select (c) **Explanation:** Unfired pressure vessels such as compressed air tanks, steam-jacketed kettles, etc. must follow the ASME code, section VIII, division 1 and 2.

Question 61

Question: The recommended maximum allowable slope for a ramp is:

- a. 45 degrees
- b. 25 degrees
- c. 20 degrees
- d. 15 degrees

Correct Answer: Select (d) **Explanation:** Ramps should be constructed using the minimum possible slope. The recommended maximum value for the slope of a ramp is 15 degrees. However, the slope should never exceed 20 degrees.

Question 62

Question: An indoor warehousing operations, the preferred type of powered truck is:

- a. Liquefied petroleum gas trucks
- b. Electric trucks
- c. Diesel trucks
- d. Gasoline trucks

Correct Answer: Select (b) **Explanation:** Electric trucks do not generate carbon monoxide and are preferred over other types for indoor operations. LPG powered trucks generate considerably less carbon monoxide compared to gasoline trucks.

Question 63

Question: The most common occupational injuries resulting from manual materials handling are injuries to the:

- a. Foot
- b. Hand
- c. Eyes
- d. Back

Correct Answer: Select (d) **Explanation:** Injuries to the back are the most common type of injuries resulting from manual materials handling. Many studies indicate that about 25 percent of all worker's compensation claims account for injuries to the lower back.

Question 64

Question: Many powered materials handling vehicles have a high center of gravity. This means that:

- a. The maximum load rating of the vehicle is increased compared to an identical low center of gravity vehicle.
- b. The pressure exerted on tires is less than an identical vehicle with a low center of gravity.
- c. The gravitational force exerted on the center of the vehicle is less than that of an identical vehicle with a low center of gravity.
- d. When the vehicle operated in a turn, it has a greater potential for rollover compared to an identical vehicle with a low center of gravity.

Correct Answer: Select (d) **Explanation:** The higher the center of gravity, the greater the potential for a rollover during a turn.

Question 65

Question: One of the problems associated with safety inspections is:

- a. Regulations may not be updated
- b. Control methods may not be economical
- c. Hazards are allowed to accumulate between inspections
- d. Lost time cases cannot be detected

Correct Answer: Select (c) **Explanation:** A safety inspection usually causes a surge in safety and health activities. However, after a period of time hazards are allowed to accumulate until the next inspection.

Question 66

Question: One of the disadvantages to an external safety inspection is:

- a. Its objectivity
- b. High cost
- c. Several follow-up inspections
- d. All of the above

Correct Answer: Select (b) **Explanation:** An external safety audit is normally conducted by outside consultants hired by the organization. The major advantage of an external audit is its "objectivity." The disadvantages include lack of familiarity of the auditor with the facility and the personnel, high cost, and the possibility of no or minimum follow-up.

Question 67

Question: During a safety inspection of your facility, you have discovered an unsafe act being committed by an employee. Your best course of action is:

- a. To stop the unsafe act and instruct how the job can be done safely.
- b. To report an unsafe act to employee's supervisor.
- c. Provide adequate training.
- d. All of the above.

Correct Answer: Select (d) **Explanation:** A sound safety management program sets forth provisions to stop unsafe acts as soon as they are discovered and to provide adequate training.

Question 68

Question: Accident causes can be classified into basic, indirect, and direct causes. An unsafe act by an employee which results in an accident is an example of which one of the following accident causes?

- a. Basic
- b. Indirect
- c. Direct
- d. Basic and direct

Correct Answer: Select (b) **Explanation:** Poor management is an example of basic causes of accidents. Unsafe acts or unsafe conditions are indirect causes of accidents. These can lead to direct causes of an accident such as unplanned release of energy which can result in fatality, injury, or property damage.

Question 69

Question: There are four different types of machine guarding: fixed, adjustable, interlocked, and self-adjusting. Which of the following is NOT an advantage of fixed guards?

- a. Can be constructed to suit many applications
- b. Can provide a high level of protection
- c. Low maintenance
- d. Adjustment and repair may require removal

Correct Answer: Select (d) **Explanation:** Statement (d) is one of the limitations of fixed guards. Another limitation is the possibility of interference with visibility.

Adjustable guards provide an adjustable barrier. One of the advantages of this type of machine guarding is that it can be constructed to suit many specific applications and stock sizes. Adjustable guards can have the disadvantage of allowing employees' hands to enter the danger zone, may require high maintenance, and they may be made ineffective by the operator.

Interlocked guards turn off the power to the machine if they are removed or pushed out of position. The openings of a self-adjusting guard are determined by the movement of the stock. As the operator moves the stock into the danger area, the guard is pushed away, providing an opening which is only large enough to admit the stock. After the stock is removed, the guard returns to the rest position.

Question 70

Question: _____ machine guards shut off or disengage power and prevent the machine from starting when the guard is opened.

- a. Fixed
- b. Adjustable
- c. Interlocked
- d. Self-adjusting

Correct Answer: Select (c) **Explanation:** Self-adjusting guards provide barriers that move according to the size of the stock entering the danger area.

Question 71

Question: OSHA requires [1910.212(a)(1)] that the employer must provide one or more methods of machine guarding to protect the operator and other employees from hazardous mechanical motion or action. Which of the following IS NOT an example of hazardous mechanical motion or action requiring machine guarding?

- a. Point of operation
- b. Ingoing nip points
- c. Rotating parts
- d. Vibrating parts

Correct Answer: Select (d) **Explanation:** The problem of vibrating parts usually does not require guarding and is solved using vibration dampers.

Question 72

Question: Which of the following is NOT true in regard to Class A hard hats?

- a. These helmets are for general services.
- b. They provide good impact protection but limited voltage protection.
- c. They are used mainly in mining, building construction, shipbuilding, lumbering, and manufacturing.
- d. They provide limited impact protection but good voltage protection.

Correct Answer: Select (d) **Explanation:** These helmets provide very limited voltage protection and are not suitable for electrical work.

Question 73

Question: Which of the following statements IS NOT true in regard to Class B hard hats?

- a. Choose Class B helmets if your employees are engaged in electrical work.
- b. They protect against falling objects and high-voltage shock and burns.
- c. The shells are water resistant and slow burning.
- d. They do not have to meet ANSI standards.

Correct Answer: Select (d) **Explanation:** Class B hard hats must meet ANSI standards. Class B hard hats fall into two different categories: ANSI Type I and ANSI Type II. Type I provides protection from electric shock and falling objects. Type II provides protection from falling and flying objects and electric shock and burns.

Question 74

Question: Which of the following statements is true in regard to Class C hard hats?

- a. They offer limited head protection.
- b. They protect against falling objects and high-voltage shock and burns.
- c. They provide good impact protection.
- d. Both (b) and (c).

Correct Answer: Select (a) **Explanation:** Class C helmets are designed for comfort. They offer limited head protection. They protect workers against fixed objects but do not protect against falling objects or electric shock.

Question 75

Question: Which of the following colors are used on the coverings of grounded conductors?

- a. Red
- b. White or natural gray
- c. Green
- d. Yellow

Correct Answer: Select (b) **Explanation:** Conductors that are grounded have a white or natural gray covering.

Question 76

Question: Many combustible gas devices burn the contaminants catalytically. This causes an increase in temperature, resulting in an increase in the resistance of the filament. Which of the following BEST describes the principle used in these devices?

- a. Wheatstone bridge
- b. Catburner
- c. Combustion chamber
- d. Catcomb

Correct Answer: Select (a) **Explanation:** The instruments are built on the Wheatstone bridge principle. As the temperature increases, the resistance of the filament also increases, causing a decrease in current flow.

Question 77

Question: Which of the following statements in NOT true in regard to cathodic protection?

- a. It is used to control the corrosion of a metal surface
- b. It makes the metal surface the cathode of an electrochemical cell
- c. It makes the metal surface the anode of an electrochemical cell
- d. It connects protected metal to a more easily corroded sacrificial metal acting as the anode

Correct Answer: Select (c) **Explanation:** In this technique, the sacrificial metal corrodes instead of the protected metal. One of the major uses of cathodic protection is protecting underground storage tanks against corrosion.

Question 78

Question: Which of the following statements is NOT true regarding a ground fault circuit interrupter (GFCI)?

- a. It is a fast-acting circuit breaker designed to shut off electric power in the event of a ground fault.
- b. It can shut off power in as little as 1/40 of a second.
- c. It works by comparing the amount of current going to and returning from the equipment along circuit conductors.
- d. It protects the user from line contact hazards.

Correct Answer: Select (d) **Explanation:** GFCI does not protect against line to line contact. For example, GFCI does not protect a person holding two hot wires, a hot and a neutral wire in each hand, or contacting an overhead power line.

Question 79

Question: CAD is an acronym for:

- a. Construction and design
- b. Cumulative anxiety disorders
- c. Computer aided design
- d. Construction and development

Correct Answer: Select (c) **Explanation:** CAD is an acronym for Computer Aided Design. CAD systems are generally comprised of both software and hardware. This combination allows the design of everything from a bridge to a ship. Generally, CAD systems also require a high quality monitor to display graphics. Nowadays, CAD software is readily available for purchase.

Question 80

Question: The two most common software packages used in occupational health and safety departments are:

- a. Industrial hygiene and occupational safety
- b. Occupational safety and regulatory compliance
- c. Word processing and database management
- d. Industrial hygiene and regulatory compliance

Correct Answer: Select (c) **Explanation:** Word processing and database management is the two most commonly used software in occupational safety and health departments.

Question 81

Question: The principle of receiving value (i.e. quality, safety, etc.) for an original purchase of a product is called:

- a. Caveat emptor
- b. Caveat venditor
- c. Caveat privitor
- d. Caveat vendor

Correct Answer: Select (b) **Explanation:** The old concept was called caveat emptor or "buyer beware." By statutory and common laws, this principle turned into caveat venditor or "seller beware."

Question 82

Question: If a product fails to meet a warranty, then:

- a. The manufacturer must be negligent
- b. The manufacturer can be held liable
- c. The consumer must prove unreasonable risk to receive compensation
- d. The consumer must prove negligence in design to receive compensation

Correct Answer: Select (b) **Explanation:** In this case the manufacturer can be held liable. All consumer products are also covered under the concept of strict liability. This means that the manufacturer is responsible for damages caused as a result of a defect in his products; no proof of negligence on the part of the consumer is needed.

Question 83

Question: A retailer has made an oral promise concerning a product's life expectancy. Which one of the following best describes this concept?

- a. Warranty of use
- b. Warranty of merchantability
- c. Express warranty
- d. Implied warranty

Correct Answer: Select (c) **Explanation:** An express warranty exists when statements in regard to certain performance characteristics of a product are made either in writing or orally.

Question 84

Question: You are a CSP working for a midsize company. During a routine inspection, you observe a few minor infractions. Which of the following courses of action would you undertake?

- a. Talk to the employees involved and instruct them on how to do the job safely.
- b. Issue a written warning to employees involved and send a copy to the Human Resources department.
- c. Notify the supervisor in writing.
- d. Issue a written warning to the employees involved and send a copy to the supervisor.

Correct Answer: Select (c) **Explanation:** It must be understood that the role of a CSP within an organization is advisory in nature both to management and staff. In this case, the best course of action is to notify the supervisor of the infractions and request for corrective action.

Question 85

Question: In database systems design, SQL is an acronym for:

- a. Secondary Quantitative Language
- b. Secondary Qualitative Language
- c. Simulation Quantitative Language
- d. Structured Query Language

Correct Answer: Select (d) **Explanation:** SQL is an acronym for Structured Query Language. SQL is a programming language that is commonly used to request, add, update, or delete data in a database.

Question 86

Question: CAD/CAM is an acronym for which of the following?

- a. Computer Aided Drawing/Camera Accepted Method
- b. Computer Aided Design/Computer Aided Manufacturing
- c. Computer Acceptable Design/Computer Aided Management
- d. Computer Acceptable Design/Computer Aided Manufacturing

Correct Answer: Select (b) **Explanation:** CAD/CAM is an acronym for Computer Aided Design/Computer Aided Manufacturing.

Question 87

Question: Which of the following statements is incorrect?

- a. The internet is a network of public and private networks.
- b. The intranet is a private network that is generally inaccessible to outsiders.
- c. The extranet is an intranet, which allows access to outsiders who are permitted.
- d. In peer-to-peer computer networks, the capability of the workstations is generally the same.

Correct Answer: Select (d) **Explanation:** Statements (a), (b), and (c) are correct. Statement (d) is not.

The Internet is a worldwide system of interconnected computers that serves private, public, academic, business, and government networks.

An intranet is a private network that usually belongs to a company. An intranet's site looks and acts like any other website, except that the firewall surrounding it provides access to only authorized individuals, usually the company's employees.

An extranet, like an intranet, is a private network. It securely shares information with other parties such as a company's customers and vendors. An extranet can be looked at as part of a company's intranet in that it provides access to authorized users outside the company.

Question 88

Question: In order to find something in the World Wide Web, one must know its URL. URL is an acronym for:

- a. Unified Reset Language
- b. Uniform Resource Language
- c. Unified Resource Language
- d. Uniform Resource Locator

Correct Answer: Select (d) **Explanation:** URL is an acronym for Uniform Resource Locator. This is a special code that tells the browser what kind of site you are visiting. All URLs start with http://.

URLs are also known as web addresses. In most browsers they are displayed at the top of the page inside an address bar.

Question 89

Question: The address of web sites starts with HTTP. This is an acronym for:

- a. High Transfer Terminal Program
- b. High Transfer Terminal Protocol
- c. Hyper Text Transport Protocol
- d. Hyper Text Transfer Program

Correct Answer: Select (c) **Explanation:** HTTP is an acronym for Hyper Text Transport Protocol. HTTP is an application protocol that is used for distributed hypermedia information systems. This protocol is the most critical in communicating data on the World Wide Web.

Question 90

Question: The most basic features of a web browser include the capability to display all of the following EXCEPT:

- a. Hypertext links
- b. Images
- c. RealPlayer by RealNetworks
- d. Text

Correct Answer: Select (c) **Explanation:** RealNetworks RealPlayer is a plug-in that can be downloaded from the Internet and be used with the web browser. It plays audio and video content in RealMedia format.

Question 91

Question: Which of the following statements is true regarding Adobe Acrobat, which can be downloaded from the Internet?

- a. Plays flash presentations, containing animations and vector graphics
- b. Displays files in its native Portable Document Format (PDF)
- c. Plays macromedia graphics and multimedia audio
- d. Plays audio and video content in RealMedia format

Correct Answer: Select (b) **Explanation:** Statement (a) is a characteristic of Adobe Flash Player. Statement (c) is a characteristic of Adobe Shockwave, and statement (d) is a characteristic of RealNetworks RealPlayer.

Question 92

Question: Which of the following software can be used as a tool for reading and composing E-mail?

- a. SeaMonkey Composer
- b. Microsoft Outlook Express
- c. Mozilla Firefox
- d. Microsoft Meeting Space

Correct Answer: Select (b) **Explanation:** SeaMonkey Composer is a simple HTML editor for creating and modifying web pages. Mozilla Firefox allows you to display web pages and a variety of web content. Microsoft Meeting Space lets you connect to audio, video, and chat conferences on the Internet.

Question 93

Question: Most E-mail programs give the option of creating mail in either plain text or HTML formats. The advantage of creating mail in HTML format is that one can add:

- a. Images
- b. Color
- c. Text formatting
- d. All of the above

Correct Answer: Select (d) **Explanation:** (a), (b), and (c) are all advantages of creating E-mail in HTML. Compared to plain text, HTML can be customized much more to personal or business needs. This is a major advantage because it provides the capability of conveying pictures, colors, emphasized text, and quotations.

Question 94

Question: UNIX workstations are generally used for which of the following applications?

- a. Low CPU applications
- b. Browser applications
- c. Hub applications
- d. CPU-intensive applications

Correct Answer: Select (d) **Explanation:** UNIX workstations are generally used for CPU-intensive applications, such as special effects and graphics.

Question 95

Question: Computer networks fall into two general categories: peer-to-peer networks and client/server networks. All of the following are considered to be advantages of a peer-to-peer network EXCEPT:

- a. Inexpensive.
- b. Simplicity.
- c. All workstations can share the same printer.
- d. The control is not centralized.

Correct Answer: Select (d) **Explanation:** Economy, simplicity, and sharing of hardware are all advantages of peer-to-peer networks. Lack of centralized control is considered to be a disadvantage of this type of networks.

Question 96

Question: All of the following statements are correct in regard to a spectrophotometer EXCEPT:

- a. It can measure the transmission of liquids or solids as a function of wavelength.
- b. It can measure absorption of liquids or solids as a function of wavelength.
- c. The beam can be split in two: one beam that goes through a reference cell, and another beam that goes through a sample cell.
- d. It measures the amount of ionizing radiation as a function of the intensity of radiation.

Correct Answer: Select (d) **Explanation:** Statements (a), (b), and (c) are true in regard to a spectrophotometer. This instrument is not used for measuring ionizing radiation.

Spectrophotometers measure the amount of light having a certain wavelength that passes through a medium. The amount of light absorbed is a measure of the concentration of solute present. For example, the concentration of a colored solute in a solution can be determined by the amount of light at a certain wavelength that is absorbed.

Question 97

Question: Many web servers and file servers run on UNIX operating systems. Which of the following statements best describes this operating system?

- a. UNIX is a Windows-based program developed by the Microsoft Corporation.
- b. UNIX is an operating system by which your network is physically laid out and developed by Procom Technology.
- c. UNIX is a multi-user, multi-tasking computer operating system. This operating system was developed by AT&T Corporation.
- d. UNIX is a protocol that allows workstations to access Internet electronic centralized servers. This operating system was developed by the XP Corporation.

Correct Answer: Select (c) **Explanation:** Statement (c) is a correct description of UNIX operating system.

An operating system allows interaction with all the software and hardware on the computer. With a graphical user interface (GUI) such as Windows, the interaction with the computer is accomplished by using the keyboard and mouse. With Unix operating systems, the user has the option of using command lines as well as the GUI.

Unix operating systems in general provide more flexibility and control as compared to Windows operating systems.

Question 98

Question: The frequency of eye injuries in a machine shop with new machines added has been on the rise. Which of the following course of actions is the MOST appropriate?

- a. Provide employees with safety glasses with side shields.
- b. Provide employees with face shields.
- c. Provide employees with face shields to be worn on top of safety glasses with side shields.
- d. Apply engineering controls and ensure that the needed personal protective equipment is used.

Correct Answer: Select (d) **Explanation:** Every attempt must be made to solve safety problems by using engineering and/or administrative controls. However, if engineering and/or administrative controls cannot solve the problem, appropriate personal protective equipment that has been matched to the hazards must be selected and used.

Question 99

Question: Use of Ground-Fault Circuit Interrupters (GFCI) is an example of which of the following?

- a. Loss reduction
- b. Loss prevention
- c. Risk reduction
- d. All of the above

Correct Answer: Select (b) **Explanation:** Use of a GFCI is a good example of engineering control for loss prevention. Loss reduction refers to situations where a loss has already occurred and the measures try to minimize further losses. For example, when emergency personnel respond to an emergency, loss has occurred and the emergency personnel try to minimize further losses.

Question 100

Question: According to OSHA regulations contained in 29 CFR 1910.145, all of the following are examples of safety signs EXCEPT:

- a. Danger signs
- b. Signs for pedestrian crossing on a public road
- c. Caution signs
- d. Safety instruction signs

Correct Answer: Select (b) **Explanation:** Signs for pedestrian crossing on a public road are DOT signs, not OSHA safety signs.

Question 101

Question: Workers in many different occupations are exposed to hexavalent chromium. Calcium chromate, chromium trioxide, lead chromate, strontium chromate, and zinc chromate are known human carcinogens. Which of the following types of cancer is associated with overexposure to hexavalent chromium?

- a. Lung cancer
- b. Skin cancer
- c. Stomach cancer
- d. Colon cancer

Correct Answer: Select (a) **Explanation:** An increase in incidence of lung cancer has been observed among workers in industries that produce chromate and manufacture pigments containing chromate. Measurement of chromium in urine is used to determine the extent of exposure to chromium. Occupational exposures occur mainly among workers who handle pigments containing dry chromate, spray paints and coatings containing chromate, operate chrome plating baths, and weld or cut metals containing chromium, such as stainless steel. Stainless steel welding involves the greatest exposure to hexavalent chromium.

Question 102

Question: Which of the following organizations issues health effects certification for tiles and installation materials?

- a. Consumer Products Safety Commission (CPSC)
- b. Underwriter's Laboratories (UL)
- c. National Institute for Occupational Safety and Health (NIOSH)
- d. National Fire Protection Association (NFPA)

Correct Answer: Select (b) **Explanation:** Underwriter's laboratories issue the certification. UL was founded in 1894 by William Henry Merrill. Today, UL provides a wide range of services such as: certification, testing, inspection, and auditing to mention a few.

Question 103

Question: Which of the following is NOT a characteristic of nanoparticles that are related to adverse health effects:

- a. Size
- b. Chemical composition and surface character
- c. Shape
- d. Color

Correct Answer: Select (d) **Explanation:** Nanoparticles are able to cross cell membranes and reach the blood and various organs because of their very small size. The degree of toxicity posed by nanoparticles depends on their chemical composition and on the composition of any chemicals absorbed onto their surfaces. Although there is little definitive evidence, the health effects of nanoparticles are likely to also depend on their shape.

Question 104

Question: Which of the following can become combustible dust under specific situations?

- a. Rubber
- b. Plastics
- c. Aluminum
- d. All of the above

Correct Answer: Select (d) **Explanation:** A combustible dust is defined as any fine material that has the potential to catch fire and explode when mixed with air at proper concentrations. Some of these materials, such as aluminum, are not combustible under normal conditions, but they can catch fire and explode if the particles are the right size and in the right concentration.

Question 105

Question: All of the following are examples of workplaces and industries that have potential for dust explosions EXCEPT:

- a. Grain elevators
- b. Metal processing
- c. Food production
- d. Crude oil distillation

Correct Answer: Select (d) **Explanation:** Dusts can be created during transportation and handling, processing, polishing, and grinding and shaping. Dust explosion is of special concern in grain elevators, metal processing operations, and some food industries.

Question 106

Question: As compared to a normal fire, dust explosion needs two additional elements known as the dust pentagon. Which of the following is one of these two additional elements?

- a. Fuel to burn
- b. Ignition Source
- c. Suspension of Particles in air
- d. Oxygen

Correct Answer: Select (c) **Explanation:** Dispersion or suspension of particles in air and confinement of the dust cloud are two additional elements required for a dust cloud explosion.

Question 107

Question: The most important concept in selection of Personal Protective Equipment (PPE) is to:

- a. Train workers in its proper use
- b. Decide on procedures for disinfection
- c. Match PPE to the hazard
- d. Ensure that it is properly stored

Correct Answer: Select (c) **Explanation:** Matching PPE to the hazard is the most important step in the selection process. Using the wrong kind of PPE can create a false sense of security and increase the probability of an accident.

Question 108

Question: OSHA requires (29CFR1910.138) that whenever an employee uses his/her own PPE:

- a. The employer is responsible to ensure that it is of the approved type.
- b. The employee assumes full responsibility.
- c. The employee assumes responsibility for PPE being of the approved type but employer is responsible for its maintenance and disinfection.
- d. OSHA requires that no employee owned PPE be used in the workplace.

Correct Answer: Select (a) **Explanation:** When employee-owned PPE is used in the workplace, the employer is responsible for ensuring that it is safe and of the approved type. It is also the responsibility of the employer to establish adequate measures for maintenance and disinfection of PPE.

Question 109

Question: Workers who are assigned to jobs requiring eye protection and wear prescription glasses must be provided with which of the following?

- a. Prescription safety glasses
- b. Prescription safety goggles
- c. Safety goggles that would fit over prescription glasses
- d. Any of the above.

Correct Answer: Select (d) **Explanation:** Any of the actions stated would satisfy the eye protection requirement.

Question 110

Question: Which of the following organizations provide recommended levels of illumination for different work areas?

- a. ANSI
- b. Illuminating Engineering Society (IES)
- c. American Society of Lighting Engineers (ASLE)
- d. NFPA

Correct Answer: Select (b) **Explanation:** The Illuminating Engineering Society (IES) provides recommended values for lighting in different industrial work areas.

Question 111

Question: Goggles with indirect ventilation ports are most appropriate for which of the following situations?

- a. Welding operations
- b. Corrosive chemicals
- c. Flying objects
- d. Cutting operations

Correct Answer: Select (b) **Explanation:** Goggles with indirect ventilation ports do not allow entry of liquids and are suitable when working with corrosive chemicals.

Question 112

Question: Hard hats must meet the requirements and specification established by which of the following organizations?

- a. NIOSH
- b. CPSC
- c. ANSI
- d. OSHRC

Correct Answer: Select (c) **Explanation:** Hard hats must meet the requirements and specifications established by ANSI in Z89.1. These specifications have been adopted by OSHA.

Question 113

Question: Which of the following statements is correct regarding a respirator wearer in an atmosphere which is considered to be Immediately Dangerous to Life or Health (IDLH)?

- a. The buddy system must be used.
- b. The worker must be supplied with an air purifying respirator.
- c. A closed loop SCBA with constant flow regulator is required.
- d. An open loop SCBA with variable flow regulator is required.

Correct Answer: Select (a) **Explanation:** In IDLH atmospheres buddy system must be used. This means that the respirator wearer must be watched by another person who is in a position to provide help in an emergency.

Question 114

Question: Air purifying respirators remove the contaminants from air primarily by which of the following mechanisms?

- a. Absorption
- b. Gas-Liquid extraction
- c. Adsorption
- d. Only (a) and (c) above.

Correct Answer: Select (c) **Explanation:** Air purifying respirators remove the contaminants from air on the bed of a solid. Removal of a component of a gaseous mixture on a solid is called adsorption. "Absorption" refers to the process of removing one or more components of a gas mixture into a liquid.

Question 115

Question: Air purifying respirators must not be used in which of the following situations?

- a. Atmospheres containing less than 19.5% oxygen by volume
- b. In IDLH atmospheres
- c. In atmospheres where contaminants cannot be detected by odor or taste at levels below PEL or TLV
- d. All of the above.

Correct Answer: Select (d) **Explanation:** Air purifying respirators do not have their own oxygen source and should not be used in oxygen deficient atmospheres (less than 19.5%). Breakthrough will occur very quickly in an IDLH atmosphere. In the absence of test data, a cartridge respirator cannot be used in an atmosphere where the contaminant of interest does not have a warning property such as odor or taste which allows the user to know that breakthrough has taken place. The odor threshold must be lower than PEL or Action Level.

Question 116

Question: Which of the following statements is true regarding a Self-Contained Breathing Apparatus (SCBA)?

- a. Open circuit SCBA's have a relatively long operating time as compared to closed circuit SCBA's.
- b. Open circuit SCBA's remove CO2 and water from exhaled air.
- c. Positive pressure SCBA's should be used in IDLH conditions.

- d. Closed circuit negative pressure SCBA provide the highest level of protection in IDLH conditions.

Correct Answer: Select (c) **Explanation:** Positive pressure SCBA's provide a positive pressure inside the face mask at all times. This would eliminate the possibility of slightest leaks from the contaminated atmosphere into the face mask. This type of respirator is most effective in atmospheres which are considered to be Immediately Dangerous to Life or Health (IDLH) or oxygen deficient.

Question 117

Question: Which of the following test methods can be used to check the fit of a respirator?

- a. Qualitative or quantitative
- b. Closed loop, open loop
- c. Positive pressure, negative pressure
- d. Only (a) and (c) above.

Correct Answer: Select (d) **Explanation:** In qualitative test a harmless agent is injected into the air and the user uses his/her sense to detect any leaks. In a quantitative test, an instrument is placed inside face-mask which actually measures the amount of leakage. In positive pressure, the user exhales into the face-mask and it should remain bulged; in negative pressure the face mask should remain collapsed when inhalation takes place.

Question 118

Question: According to OSHA regulations, a Self-Contained Breathing Apparatus (SCBA) must be inspected at least once every _____ days and after each use:

- a. 15 days
- b. 30 days
- c. 7 days
- d. 45 days

Correct Answer: Select (b) **Explanation:** OSHA requires that SCBA's be inspected at least once every 30 days and after each use (29CFR 1910.133).

Question 119

Question: Butyl rubber has a solubility parameter of 16. Gloves made of butyl rubber are most effective against which of the following chemicals.

- a. A chemical with a solubility parameter of 18
- b. A chemical with a solubility parameter of 28
- c. A chemical with a solubility parameter of 16
- d. A chemical with a solubility parameter of 47

Correct Answer: Select (d) **Explanation:** The concept of solubility parameter is the extension of the old rule that "like dissolves like." The closer the values of the solubility parameters of two chemicals, the more soluble they are in each other. In this case, a chemical with solubility parameter of 47 is the least soluble in butyl rubber with a solubility parameter of 16.

Question 120

Question: OSHA's Process Safety Management standard applies to all of the following EXCEPT:

- a. A process which involves a chemical at or above specified threshold quantities.
- b. A process which involves a flammable liquid on site in one location, in a quantity of 10,000 lb or more.
- c. Hydrocarbon fuels used solely for workplace consumption for fuel.
- d. A process which involves a flammable gas on site in one location, in a quantity of 10,000 lb or more.

Correct Answer: Select (c) **Explanation:** Hydrocarbon fuels such as propane used solely for workplace consumption are exempt from this OSHA standard.

Question 121

Question: The Process Safety Management of highly hazardous chemicals is contained in:

- a. 40 CFR part 263
- b. 29 CFR 1910.1200
- c. 29 CFR 1910.120
- d. 29 CFR 1910.119

Correct Answer: Select (d) **Explanation:** The process safety management standard is contained in 29 CFR 1910.119. Appendix C of 29 CFR 1910.119 contains non-mandatory compliance for process safety management.

Question 122

Question: All of the following are exempt from Process Safety Management standard EXCEPT:

- a. Normally unoccupied remote facilities.
- b. Retail facilities.
- c. Oil or gas well-drilling
- d. A process, which involves a flammable gas in one location in quantities of 10,000 lb or more.

Correct Answer: Select (d) **Explanation:** This OSHA standard does not apply to retail facilities, oil or gas well-drilling or servicing operations, or normally unoccupied remote facilities. This standard applies to any process which involves a flammable liquid or gas, as defined in OSHA's Hazard Communication Standard in 29 CFR 1910.1200 (c), on site in one location, in a quantity of 10,000 lb or more.

Question 123

Question: Under PSM standard an "atmospheric tank" means:

- a. A tank, which is designed to operate from atmospheric pressure to 0.5 psig.
- b. A tank, which is designed to operate under atmospheric pressure.
- c. A tank, which is designed to operate at or below atmospheric pressures.
- d. A tank, which is designed to operate between atmospheric pressure and 28 psig.

Correct Answer: Select (a) **Explanation:** "Atmospheric tank" means a storage tank, which has been designed to operate at pressures from atmospheric through 0.5 psig (pounds per square inch gauge). It must be pointed out that pressure can be expressed in two different scales, absolute pressure scale and gauge pressure scale. The relationship between these scales is: Absolute pressure = Gauge pressure + Atmospheric pressure

---

Question 124

Question: The term "hot work" under PSM standard includes all of the following operations EXCEPT:

- a. Electric.
- b. Welding.
- c. Brazing.
- d. Any operation, which involves temperatures exceeding 1000 degrees Celsius.

Correct Answer: Select (d) **Explanation:** The term "hot work" means work involving electric or gas welding, cutting, brazing, or similar flame or spark-producing operations.

Question 125

Question: For existing equipment designed and constructed many years ago with the codes and standard available at that time and no longer in general use, as part of the "process safety information" package of the PSM standard, the employer must do all of the following EXCEPT:

- a. Document the codes and standards that were used.
- b. Change the design or install new equipment in compliance with present codes and standards
- c. Document that the design and construction are still suitable for the intended use.
- d. Document that the testing, inspection and operation are still suitable for the intended use.

Correct Answer: Select (b) **Explanation:** The information pertaining to process equipment design must be documented. In other words, what were the codes and standards relied on to establish good engineering practice.

For existing equipment designed and constructed many years ago in accordance with the codes and standards available at that time and no longer in use today, the employer must document which codes and standards were used and that the design and construction along with the testing, inspection and operation are still suitable for the intended use. Where the process technology requires a design which departs from the applicable codes and standards, the employer must document that the design and construction is suitable for the intended purpose.

Question 126

Question: Selection of a "process hazard analysis" methodology or technique will be influenced by all of the following factors EXCEPT:

- a. The amount of existing knowledge about the process.
- b. The size and complexity of the process.
- c. Maintenance procedures.
- d. Is it a new process or one which has been changed frequently by the inclusion of innovative features?

Correct Answer: Select (c) **Explanation:** Selection of a PHA methodology or technique will be influenced by many factors including: • The amount of existing knowledge about the process. • Is it a process that has been operated for

a long period of time with little or no innovation and extensive experience has been generated with its use? • Is it a new process or one which has been changed frequently by the inclusion of innovative features? • The size and complexity of the process.

Question 127

Question: Process chemicals information is part of the "process safety information" package. Which of the following IS NOT expected to be included in the process chemicals information?

- a. Erosion effects.
- b. Explosion characteristics.
- c. Spill cleanup procedures
- d. Corrosion effects

Correct Answer: Select (c) **Explanation:** The information to be compiled about the chemicals, including process intermediates, needs to be comprehensive enough for an accurate assessment of the fire and explosion characteristics, reactivity hazards, the safety and health hazards to workers, and the corrosion and erosion effects on the process equipment and monitoring tools. Current safety data sheet (SDS) information can be used to help meet this requirement which must be supplemented with process chemistry information including runaway reaction and over pressure hazards if applicable.

Question 128

Question: Under "process technology information" part of the "process safety information" package, employers are encouraged to use diagrams, which help users understand the process. Which of the following IS NOT one of the three general categories of the recommended diagrams?

- a. Block flow diagrams.
- b. Fault tree diagrams.
- c. Process flow diagrams.
- d. Piping and instrumentation diagrams.

Correct Answer: Select (b) **Explanation:** The 3 general categories of recommended diagrams are BLOCK Flow Diagrams (BFD), Process Flow Diagrams (PFD), and Piping and Instrumentation Diagrams (P&IDs).

Question 129

Question: OSHA's process safety management standard does not apply to "normally unoccupied remote facilities." Which of the following statements BEST describes this term?

- a. A facility which is at least 80- miles away from all cities with a population of 5,000 or more.
- b. A facility, which is operated, maintained or serviced by employees who visit the facility only periodically to check its operation and to perform necessary operating or maintenance tasks.
- c. A facility which is operated, maintained or serviced by employees who visit the facility no more than once every 6 months to check its operation and to perform necessary operating or maintenance tasks.
- d. All of the above.

Correct Answer: Select (b) **Explanation:** "Normally unoccupied remote facility" means a facility which is operated, maintained or serviced by employees who visit the facility only periodically to check its operation and to perform

necessary operating or maintenance tasks. No employees are permanently stationed at the facility. Facilities meeting this definition are not contiguous with, and must be geographically remote from all other buildings, processes or persons.

Question 130

Question: Under OSHA's process safety management standard, the term "highly hazardous chemical" means:

- a. A substance possessing toxic, reactive, flammable, or explosive properties and specified by the standard.
- b. A substance possessing toxic, reactive, flammable, or explosive properties and specified by the standard and also listed under TSCA.
- c. Toxic chemicals with an LD50 of less than 5000 mg/kg, which are specified by the standard.
- d. Hazardous chemicals listed by OSHA under Hazard Communication Standard (29 CFR 1910.1200).

Correct Answer: Select (a) **Explanation:** "Highly hazardous chemical" means a substance possessing toxic, reactive, flammable, or explosive properties and specified by this standard.

Question 131

Question: Which of the following IS NOT an element of process safety management as defined by OSHA?

- a. Compliance audits.
- b. Trade secrets
- c. Hazardous waste management.
- d. Contractors.

Correct Answer: Select (c) **Explanation:** The 14 Elements of process safety management as defined by OSHA are listed below. Hazardous waste management is not an element of the process safety management standard. (a) Employee participation. (b) Process safety information. (c) Process hazard analysis. (d) Operating procedures. (e) Training. (f) Contractors. (g) Pre-startup safety review. (h) Mechanical integrity. (i) Hot work permit. (j) Management of change. (k) Incident investigation. (l) Emergency planning and response. (m) Compliance audits. (n) Trade secrets.

Question 132

Question: The "employee participation" element of the process safety management standard is required under which of the following legislative acts?

- a. CERCLA.
- b. Clean Air Act Amendments.
- c. HAZWOPER.
- d. HAZCOM.

Correct Answer: Select (b) **Explanation:** Section 304 of the Clean Air Act Amendments states that employers are to consult with their employees and their representatives regarding the employer's efforts in the development and implementation of the process safety management program elements and hazard assessments. Section 304 also requires employers to train and educate their employees and to inform affected employees of the findings from incident investigations required by the process safety management program.

Question 133

Question: Which of the following IS NOT addressed in the "process safety information" element of the process safety management standard?

- a. Incident investigation.
- b. Process Technology.
- c. Process equipment.
- d. Process Chemicals.

Correct Answer: Select (a) **Explanation:** Incident investigation is a separate element of the process safety management standard and is not part of the "process safety information" package.

Question 134

Question: "Process technology information" is a component of the "process safety information" package required by the process safety management standard. Which of the following IS NOT expected to be included in the "process technology information"?

- a. Maximum inventory limits for the process chemicals.
- b. Limits beyond which would be considered upset conditions.
- c. A qualitative estimate of the consequences or results of deviations from the design intent.
- d. Determination of toxic, flammable, reactive, and explosive properties of the process chemicals.

Correct Answer: Select (d) **Explanation:** Determination of fire and explosion characteristics, reactivity hazards, the safety and health hazards to workers, and the corrosion and erosion effects on the process equipment and monitoring tools is part of the "process chemicals information" of the "process safety information" package.

Question 135

Question: The "employee training" element of the process safety management standard must cover all of the following areas EXCEPT:

- a. Operating procedures and safe work practices.
- b. How to interpret chemical labels and Safety Data Sheets.
- c. Safety procedures.
- d. Routine and non-routine work authorization activities.

Correct Answer: Select (b) **Explanation:** Interpretation of chemical labels and SDS's is required under OSHA's Hazard Communication Standard (29 CFR 1910.1200). Under provisions of the process safety management standard all employees, including maintenance and contractor employees, involved with highly hazardous chemicals need to fully understand the safety and health hazards of the chemicals and processes they work with for the protection of themselves, their fellow employees and the citizens of nearby communities. The training must cover the following areas: • Operating procedures and safe work practices. • Emergency evacuation and response. • Safety procedures. • Routine and non-routine work authorization activities.

Question 136

Question: The training program required under the "employee training" element of the process safety management standard must accomplish all of the following EXCEPT:

- a. Define subjects to be covered.
- b. Develop a training center within the facility.
- c. Define employees to be trained.
- d. Define goals and objectives of the training program.

Correct Answer: Select (b) **Explanation:** Development of a training center is not a requirement of this standard. The training program must accomplish the following: • Define employees to be trained. • Subjects to be covered. • Define goals and objectives of the training program. • Establish means to determine whether the goals and objectives have been achieved.

Question 137

Question: Which of the following IS NOT true regarding the team conducting the process hazard analysis required under the process safety management standard?

- a. The team needs to understand the hazard analysis methodology that is going to be used.
- b. The team can vary in size.
- c. The team members must have varied operational and technical backgrounds.
- d. At least one member of the team must have a college degree from an accredited institution in a field related to the process.

Correct Answer: Select (d) **Explanation:** Possession of a college degree in a field related to the process is not a requirement under this standard. The team conducting the PHA, however, needs to understand the methodology that is going to be used. A PHA team can vary in size from two people to a number of people with varied operational and technical backgrounds. The team leader needs to be fully knowledgeable in the proper implementation of the PHA methodology that is to be used.

Question 138

Question: Detailed guidance on the content and application of process hazard analysis methodologies can be obtained from which of the following professional organizations?

- a. American Society of Safety Engineers (ASSE).
- b. American Institute of Chemical Engineers (AICHE).
- c. American Institute of Mechanical Engineers (ASME).
- d. American Chemical Society (ACS)

Correct Answer: Select (b) **Explanation:** The Center for Chemical Process Safety (CCPS) of the American Institute of Chemical Engineers can provide guidance on process hazard analysis methodologies.

Question 139

Question: The acronym SOP in the process safety management standard is an abbreviation for:

- a. Standard Operating Practices.
- b. Safety Officer's Program.

- c. Safe Operation of Process.
- d. Safety of Personnel.

Correct Answer: Select (a) **Explanation:** SOP is an acronym for Standard Operating Practices.

Question 140

Question: Under the "operating procedures" element of the process safety management standard, the employer is required to certify that the operating procedures are accurate and current. This certification must be done:

- a. Every 2 years.
- b. Every 3 years.
- c. Every 5 years.
- d. Annually.

Correct Answer: Select (d) **Explanation:** The certification must be done annually.

Question 141

Question: Employers who use contractors to perform work in and around processes that involve highly hazardous chemicals must do all of the following EXCEPT:

- a. Establish a screening process to hire contractors who can accomplish the tasks safely.
- b. Evaluate contractor's work methods and experiences.
- c. Ensure that the contractor employees have successfully completed the OSHA 40-hour HAZWOPER training.
- d. Maintain a site injury and illness log for the contractors.

Correct Answer: Select (c) **Explanation:** This standard does not require HAZWOPER training for contractor employees. HAZWOPER is covered in 29 CFR 1910.120.

Question 142

Question: The "pre-startup safety review" component of the process safety management standard requires all of the following EXCEPT:

- a. Construction of equipment are in accordance with design specifications.
- b. Operating procedures are in place and are adequate.
- c. A compliance audit has been conducted on the process.
- d. Training has been completed.

Correct Answer: Select (c) **Explanation:** Compliance audits are a separate element of the process safety management standard and is not required under pre-startup safety review.

Question 143

Question: Which of the following IS NOT required under the mechanical integrity program required by process safety management standard?

- a. Written procedures to maintain the ongoing integrity of process equipment.
- b. Training for process maintenance activities.
- c. Inspection and testing.
- d. Pre-startup safety review.

Correct Answer: Select (d) **Explanation:** Pre-startup safety review is a separate element of the process safety management standard and is not addressed under mechanical integrity.

Question 144

Question: The "mechanical integrity" element of the process safety management standard requires that a list of process equipment and instrumentation for inclusion in the program be developed. This list must include all of the following EXCEPT:

- a. Pressure vessels.
- b. Storage tanks.
- c. Fire protection system components.
- d. Office buildings.

Correct Answer: Select (d) **Explanation:** A list of process equipment and instrumentation for inclusion in the program must be developed. This list must include pressure vessels, storage tanks, process piping, relief and vent systems, fire protection system components, emergency shutdown systems, alarms, interlocks, and pumps. This equipment list must be prioritized to denote which pieces of equipment require closer scrutiny than others.

Question 145

Question: Determination of the MTBF for various instrumentation and equipment is helpful in creating a sound mechanical integrity program. MTBF is an acronym for:

- a. Mechanical Threshold Based on Fabrication.
- b. Mechanical Tension Because of Failure.
- c. Mean Time Between Failures.
- d. Manufacturer's Testing By Fabrication.

Correct Answer: Select (c) **Explanation:** MTBF is acronym for Mean Time Between Failures.

Question 146

Question: Which of the following IS NOT required by the "mechanical integrity" element of the process safety management standard?

- a. Investigation of incidents.
- b. Preventative maintenance training.
- c. Quality assurance.
- d. Training for process maintenance activities.

Correct Answer: Select (a) **Explanation:** Incident investigation is a separate element of process safety management and is not required under mechanical integrity program.

Question 147

Question: Which of the following statements IS NOT TRUE regarding "hot work permit" required under process safety management standard?

- a. The permit must document the date(s) authorized for hot work.
- b. The permit must identify the object on which hot work is to be performed.
- c. The permit must document that the fire prevention and protection requirements in 29 CFR 1910.152 (a) have been implemented prior to beginning the hot work operations.
- d. The permit must be kept on file for a period of 5 years

Correct Answer: Select (d) **Explanation:** The permit must only be kept on file until the completion of hot work operations.

Question 148

Question: Under the "management of change" element of the process safety management standard, the employer is responsible to implement written procedures to manage changes. These procedures must assure that all of the following issues are addressed EXCEPT:

- a. The technical basis for the proposed change.
- b. Modifications to operating procedures.
- c. Necessary time period.
- d. Effect of change on compliance audits program.

Correct Answer: Select (d) **Explanation:** Effect of change on compliance audits program is not addressed in "management of change" element of the process safety management.

Question 149

Question: Which or the following statements IS NOT TRUE regarding the "investigation of incidents" element of the process safety management standard?

- a. Incident investigation is the process of identifying and underlying the causes of incidents and implementing steps to prevent similar events from occurring.
- b. A report of the findings of the incident investigation team must be shared with the local fire department.
- c. Employers need to develop in-house capability to investigate incidents that occur in their facilities.
- d. A team needs to be assembled and trained in the techniques of incident investigation.

Correct Answer: Select (b) **Explanation:** This standard does not require that a report of findings of the incident investigation be shared with the local fire department.

Question 150

Question: The process safety management standard requires that employers develop and implement an emergency action plan which will facilitate the prompt evacuation of employees due to an unwanted release of a highly hazardous chemical. This plan must address all of the following EXCEPT:

- a. Alarm system.
- b. Process hazard analysis.
- c. Evacuation/relocation.
- d. Preplanning for releases.

Correct Answer: Select (b) **Explanation:** Process hazard analysis is a separate element of the process safety management standard and is not addressed in the emergency action plan.

Question 151

Question: Under provisions of the process safety management standard, it is the responsibility of the employer to certify that they have evaluated compliance with the provisions of this OSHA standard. This certification must be done at least every:

- a. 2 years.
- b. 4 years.
- c. 3 years.
- d. 5 years.

Correct Answer: Select (c) **Explanation:** It is the responsibility of the employers to certify that they have evaluated compliance with the provisions of this OSHA standard at least every three years to verify that the procedures and practices developed under the standard are adequate and are being followed.

Question 152

Question: Under provisions of process safety management standard, the employer is responsible for developing a report of the findings of the compliance audit. Which of the following is true in regard to compliance audit report?

- a. Employers must retain each report for a period of at least 3 years.
- b. Employers must retain each report for a period of at least 2 years.
- c. Employers must retain the two (2) most recent compliance audit reports.
- d. Employers must retain each report for a period of at least 4 years.

Correct Answer: Select (c) **Explanation:** Employers must retain the two (2) most recent compliance audit reports.

Question 153

Question: Which of the following IS NOT required by the "compliance audits" element of the process safety management standard?

- a. The compliance audit must be conducted by at least one person knowledgeable in the process.
- b. A report of findings of the audit must be developed.
- c. Employers must retain the two (2) most recent compliance audit reports.
- d. Employers must develop complete and accurate written information concerning process chemicals, process technology, and process equipment.

Correct Answer: Select (d) **Explanation:** These issues are addressed in the "process safety information" element of the process safety management standard.

Question 154

Question: Which of the following statements IS NOT TRUE in regard to "trade secrets" element of process safety management standard?

- a. Employers must make all information necessary to comply available to those persons responsible for compiling the process safety information and to those assisting in the development of the process hazard analysis without regard to possible trade secret status of such information.
- b. Employers must make all information necessary to comply available to those persons responsible for developing the operating procedures and those involved in incident investigations without regard to possible trade secret status of such information.
- c. The employer cannot require the persons to whom the information is made available to enter into confidentiality agreements not to disclose the information.
- d. Employers must make all information necessary to comply available to those persons responsible for developing emergency planning and response and compliance audits without regard to possible trade secret status of such.

Correct Answer: Select (c) **Explanation:** The employer can ask the parties involved to enter into a confidentiality agreement not to disclose the information.

Question 155

Question: Which of the following acts is considered to be "distracted driving"?

- a. Talking to a passenger.
- b. Eating.
- c. Texting.
- d. All of the above.

Correct Answer: Select (d) **Explanation:** An estimated 1 in 4 car crashes involve cell phone use. According to the United States Department of Transportation, text messaging while driving creates a crash risk 23 times higher than driving while not distracted. Despite these statistics, more than 37% of drivers have admitted to sending or receiving text messages while driving, and 18% admit doing so regularly.

Question 156

Question: The Department of Transportation requires that every commercial motor vehicle be inspected. DOT also requires that the original or copy of the inspection report be retained for a period of _____ from the date of inspection:

- a. 14 months
- b. 12 months
- c. 6 months
- d. 24 months

Correct Answer: Select (a) **Explanation:** The inspection report must be retained by the motor carrier or other entity who is responsible for the inspection for a period of 14 months from the date of inspection.

Question 157

Question: Automatic braking technology used in automobiles uses sensors to detect other vehicles or obstacles. Which of the following is NOT a type of sensor used in this technology?

- a. Video
- b. Ultrared
- c. Infrared
- d. Ultrasonic

Correct Answer: Select (b) **Explanation:** Sensors to detect other vehicles or obstacles can include radar, video, infrared, ultra-sonic or other technologies. GPS sensors can detect fixed dangers such as approaching stop signs through a location database.

Question 158

Question: Which of the following statements is NOT true in regard to Anti-lock Braking System (ABS)?

- a. ABS allows the wheels on a motor vehicle to maintain tractive contact with the road surface.
- b. ABS can significantly decrease braking distance on loose surfaces.
- c. Recent versions of ABS not only prevent wheel lock but also electronically control the front-to-rear brake bias.
- d. ABS prevents the wheels from ceasing rotation and avoids uncontrolled skidding.

Correct Answer: Select (b) **Explanation:** Anti-lock braking system (ABS) is an automobile safety system that allows the wheels on a motor vehicle to maintain tractive contact with the road surface according to driver inputs while braking, preventing the wheels from locking up (ceasing rotation) and avoiding uncontrolled skidding.

ABS generally offers improved vehicle control and decreases stopping distances on dry and slippery surfaces for many drivers; however, on loose surfaces like gravel or snow-covered pavement, ABS can significantly increase braking distance, although still improving vehicle control.

Question 159

Question: Which of the following statements is true in regard to Vehicular Accident Reconstruction (VAR)?

- a. It is the scientific process investigating, analyzing, and drawing conclusions about the causes and events during a vehicle collision.
- b. The laws of physics and engineering principles are the basis for analysis.
- c. Results from accident reconstruction can be useful in developing recommendations for making roads and highways safer.
- d. All of the above.

Correct Answer: Select (d) **Explanation:** Vehicular accident reconstruction is the scientific process of investigating, analyzing, and drawing conclusions about the causes and events during a vehicle collision. Reconstruction personnel are employed to conduct in-depth collision analysis and reconstruction to identify the collision causation

and contributing factors in different types of collisions, including the role of the driver(s), vehicle(s), roadway and the environment. The laws of physics and engineering principles such as the conservation of linear momentum, work-energy methods, and kinematics are the basis for these analysis and may make use of software to calculate useful quantities.

Results from accident reconstructions are also useful in developing recommendations for making roads and highways safer, as well as improving safety aspects of motor vehicle designs.

Question 160

Question: Pressure relief systems use which of the following to reduce or depressurize a system:

- a. Pressure measurement
- b. Comparing the measured value to a set point to generate an error
- c. Acting on a final control element to reduce the pressure or depressurize the system
- d. All of the above

Correct Answer: Select (d) **Explanation:** Any automatic control system must first measure the parameter to be controlled. The measured value is then compared to a set point value to generate an error. The generated error (deviation from the set point) acts on a final control element such as a valve which is loaded with a spring. The spring causes the valve to open in case of excessive pressure.

Question 161

Question: In order to accurately conduct noise measurements, sound level meters should be calibrated:

- a. Once a month.
- b. Before each use.
- c. After each use.
- d. Before, during, and after each use.

Correct Answer: Select (d) **Explanation:** Statement (d) ensures accurate noise measurements.

Question 162

Question: Noise level measurements in a machine shop indicate that the TWA in an 8-hour shift exceed the allowable limits. Management decides to reduce the exposure to four hours by job rotation which in turn brings the exposure down to allowable limits. This is an example of:

- a. Engineering control
- b. Management of change
- c. Administrative control
- d. All of the above

Correct Answer: Select (c) **Explanation:** Engineering controls reduce the hazards permanently, and in many cases, eliminate the need for personal protective equipment (PPE). An example of engineering control in this case would be to place the worker in an enclosure to minimize exposure to excessive noise.

Question 163

Question: Which of the following statements is true about confined spaces?

- a. Have toxic chemicals in their atmosphere
- b. Have limited means for entry or exit
- c. Must be equipped with a rescue ladder
- d. All of the above

Correct Answer: Select (b) **Explanation:** A confined space has limited or restricted means for entry or exit and is not designed for continuous occupancy. Examples of confined spaces include tanks, vessels, silos, etc.

Question 164

Question: In order to represent the real-world use of hearing protection devices, ANSI recommends that the manufacturer's claim for noise reduction:

- a. Be reduced
- b. Be increased
- c. Remains unchanged
- d. Be evaluated and adjusted

Correct Answer: Select (a) **Explanation:** ANSI recommends that the manufacturer's claim for noise reduction of earmuff hearing protection devices be reduced by 40% to represent the actual noise reduction in real-world work situations.

Question 165

Question: There are four classes of lasers. Which of the following is true for class 1 lasers?

- a. Non-hazardous under normal conditions of use
- b. Can damage unprotected skin
- c. Capable of ionizing the material through which it flows
- d. Capable of adversely affecting the DNA

Correct Answer: Select (a) **Explanation:** Class 1 lasers are considered to be non-hazardous unless viewed through magnifiers.

Question 166

Question: Which of the following personal protective equipment (PPE) should be worn when working with electricity?

- a. Safety glasses and insulated gloves & clothing.
- b. Class B hard hat and insulated gloves & clothing.
- c. Tools that are properly grounded and insulated gloves & clothing.
- d. Non-conducting ladder and insulated gloves & clothing

Correct Answer: Select (b) **Explanation:** Class B hard hats provide protection against electric shock and must meet ANSI standards. Class B hard hats and insulated gloves & clothing must be used when working with electricity. The glove rating must match with the work to be performed. Not all gloves can be used to prevent electric shock. Ladders and tools are not personal protective equipment.

Question 167

Question: Which of the following is an example of an active safety device?

- a. Automobile seat belts.
- b. Automatic brake systems.
- c. Automobile air bags.
- d. Guard rails for fall protection.

Correct Answer: Select (b) **Explanation:** Active Safety includes set of safety features which reduce the chances of an accident or collision in the first place. However, the airbags, seatbelts, and others safety features come into play during an accident. Hence, they are called the passive safety systems.

Question 168

Question: Which of the following statements is correct regarding Underwriter's Laboratories (UL)?

- a. Sets standards for different product categories and tests products to make sure they meet the standards.
- b. UL approval is required by law.
- c. UL is part of the U.S. Department of Commerce.
- d. UL is a non-profit organization.

Correct Answer: Select (a) **Explanation:** UL approval is not legally required. It's only common because many large companies will not buy equipment that has not passed UL's safety tests. UL is not part of any government agencies and is a private safety testing organization. On January 1, 2012, Underwriters Laboratories transformed from a non-profit organization to a for-profit company in the U.S.

Question 169

Question: Which of the following best describes the basic strategy of a safe system approach for road traffic safety?

- a. To ensure that in the event of a crash, the impact energies remain below the threshold likely to produce either death or serious injury.
- b. To apply fault tree analysis to identify and eliminate the events that can lead to an accident.
- c. To ensure that the airbags are properly designed and can absorb the force generated as a result of a crash.
- d. All the above.

Correct Answer: Select (a) **Explanation:** The basic strategy of a Safe System approach is to ensure that in the event of a crash, the impact energies remain below the threshold likely to produce either death or serious injury. This threshold will vary from crash scenario to crash scenario, depending upon the level of protection offered to the road users involved. For example, the chances of survival for an unprotected pedestrian hit by a vehicle diminish

rapidly at speeds greater than 30 km/h, whereas for a properly restrained motor vehicle occupant the critical impact speed is 50 km/h (for side impact crashes) and 70 km/h (for head-on crashes).